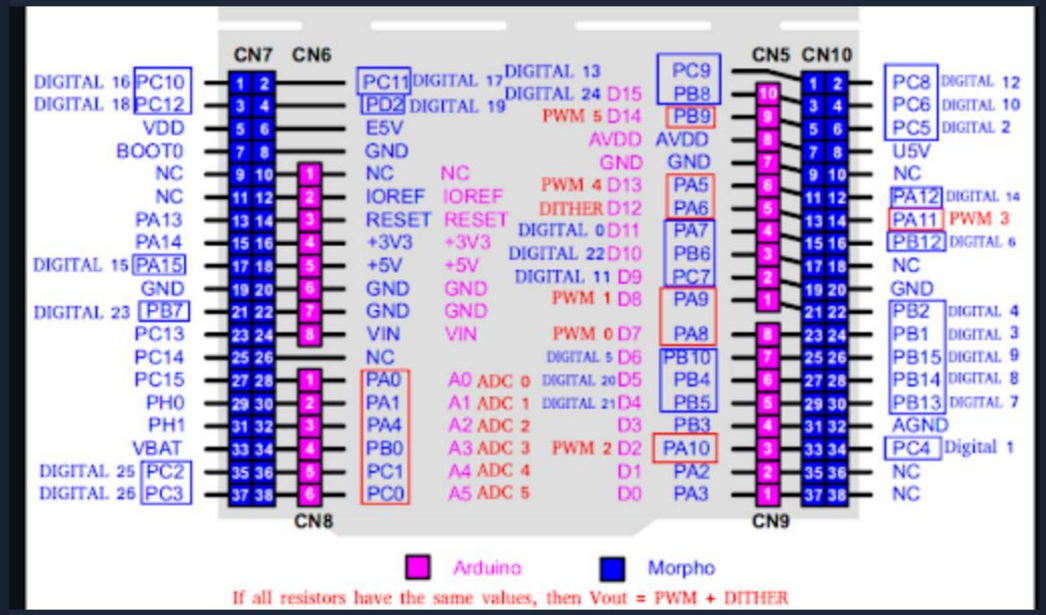
BOARD INTERCONNECT

STM32 Morpho Wiring Guide

Master-slave configurations for SPI2, I2C1, and USART1.



Base Requirements

Morpho Only

Strict blue header utilization. Zero pink Arduino pins.

Isolated Power

Independent USB cables. Isolated 3V3 and 5V lines.

Common Ground

Mandatory **`CN10-20`** linkage across both boards.

SPI (SPI2) Mapping

Direct master-to-slave pinout. Zero CS wire.

Signal	Master Pin	Slave Pin	Routing
SCK	`CN10-30 (PB13)`	`CN10-30 (PB13)`	Straight through
MOSI	`CN10-26 (PB15)`	`CN10-26 (PB15)`	Straight through
MISO	`CN10-28 (PB14)`	`CN10-28 (PB14)`	Straight through
GND	`CN10-20`	`CN10-20`	**Common link required**

I2C1 PROTOCOL

Straight pins. Breadboard pulls.

SCL Line

`CN10-3 (PB8)`

SDA Line

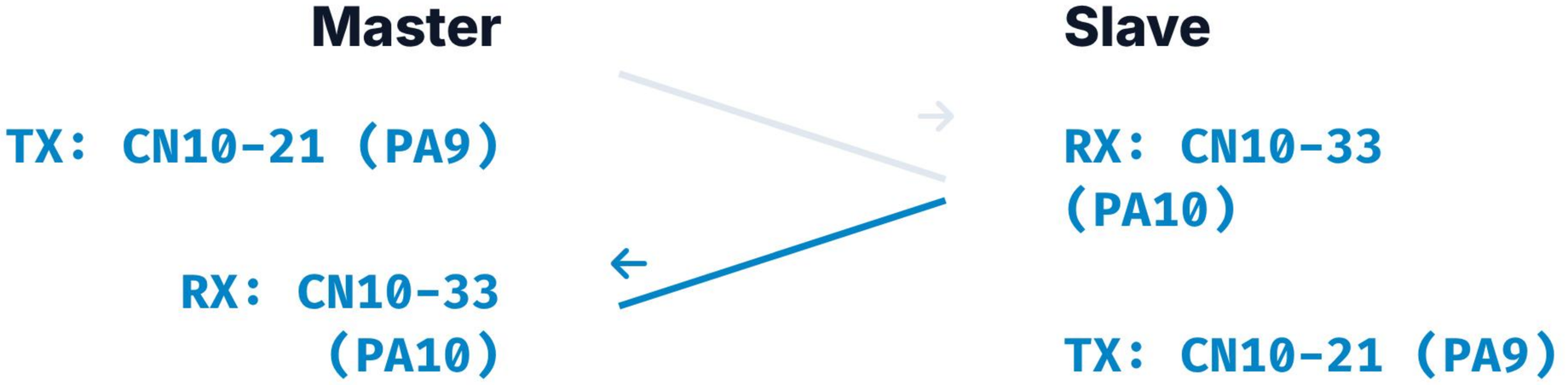
`CN10-5 (PB9)`

4.7 kΩ Pull-up Logic

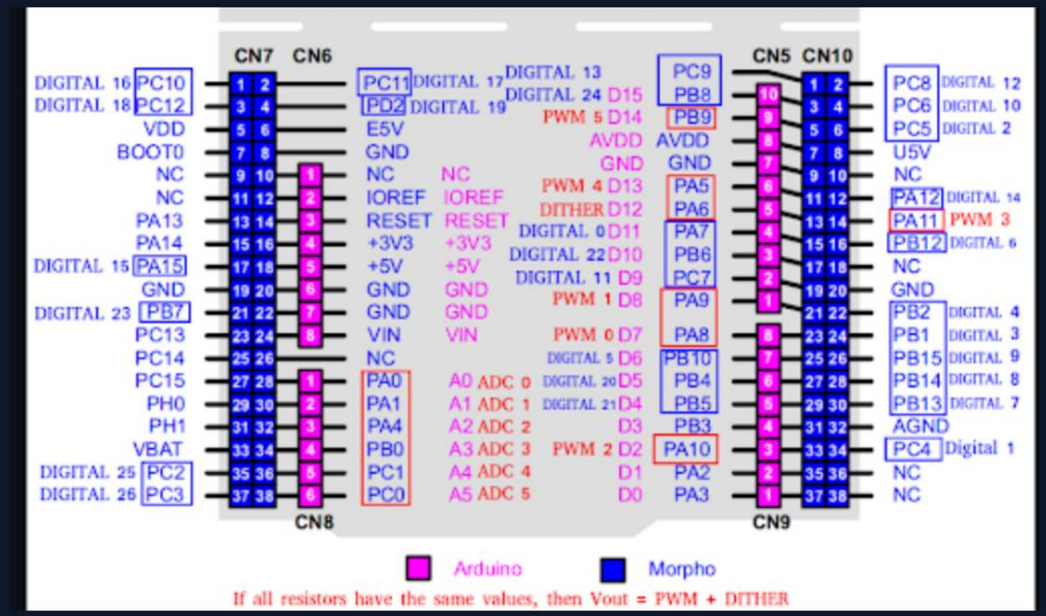
- ✓ External breadboard required.
- ✓ Resistor 1: `CN7-16 (+3V3)` to SCL.
- ✓ Resistor 2: `CN7-16 (+3V3)` to SDA.
- ⚠ **One set per bus total.**

UART (USART1) Mapping

Mandatory line crossing. Fixed register setup.



Locating CN10 Pins



ODD VS EVEN

Odd pins on left column.
Even pins on right column.

ROW CALCULATION

`Row = pin ÷ 2`

*Round up for odd numbers.